

Which Error Can the District Afford?

AP Statistics · Unit 3 · Topic 3.8 · TEACHER KEY

CED TETHER

- **Skill 1.B** — Identify key and relevant information to answer a question or solve a problem.
- **Skill 3.A** — Determine relative frequencies, proportions, or probabilities using simulation or calculations.
- **Skill 4.A** — Make an appropriate claim or draw an appropriate conclusion.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU UNC-5** — Probabilities of Type I and Type II errors influence inference.
- **LO UNC-5.A** — Identify Type I and Type II errors. [Skill 1.B]
- **EK UNC-5.A.1** — A Type I error occurs when the null hypothesis is true and is rejected (false positive).
- **EK UNC-5.A.2** — A Type II error occurs when the null hypothesis is false and is not rejected (false negative).
- Table of Errors:
 - H_0 true
 - H_a true
 - Reject H_0
 - Type I Error
 - Correct Decision
 - Fail to Reject H_0
 - Correct Decision
 - Type II Error
- **LO UNC-5.B** — Calculate the probability of a Type I and Type II errors. [Skill 3.A]
- **EK UNC-5.B.1** — The significance level, α , is the probability of making a Type I error, if the null hypothesis is true.
- **EK UNC-5.B.2** — The power of a test is the probability that a test will correctly reject a false null hypothesis.
- **EK UNC-5.B.3** — The probability of making a Type II error = 1 - power.
- **LO UNC-5.C** — Identify factors that affect the probability of errors in significance testing. [Skill 4.A]
- **EK UNC-5.C.1** — The probability of a Type II error decreases when any of the following occurs, provided the others do not change: i. Sample size(s) increases.
- ii. Significance level (α) of a test increases.
- iii. Standard error decreases.
- iv. True parameter value is farther from the null.
- **LO UNC-5.D** — Interpret Type I and Type II errors. [Skill 4.B]
- **EK UNC-5.D.1** — Whether a Type I or a Type II error is more consequential depends upon the situation.
- **EK UNC-5.D.2** — Since the significance level, α , is the probability of a Type I error, the consequences of a Type I error influence decisions about a significance level.

Essential question: How should the consequences of false positives and false negatives shape a testing plan?

DOK-3 skill: 4.B · **FRQ pattern:** alpha-vs-power-which-error-tradeoff-fits-consequences

DOK rationale: Part (c) weighs both conditional error risks and their consequences to recommend a plan for a stated priority and explain why another priority can support another plan.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- If $p = 0.10$, what decision creates the dangerous miss?
- What does the complement of 0.748 represent?
- What does Plan B trade per 1,000 tests?

[WARN] LOOK FOR

- Using $1 - \alpha$ as the probability of a Type II error.
- Claiming that lowering alpha lowers both error probabilities when n is fixed.
- Choosing solely from the larger power without acknowledging its false-positive cost.

SCORING PART (c) — E / P / I

Expected elements

- **judgment-1** (required) — Makes an office-linked recommendation and uses the two Type I risks, 0.010 and 0.050.
- **judgment-2** (required) — Uses Type II risks 0.252 and 0.123 and connects each error type to keeping or replacing the system in the appropriate true state.
- **judgment-3** (required) — Critiques both absolute claims and explains why different priorities can justify different choices.

E	Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.
P	Shows at least one substantive correct check, but has an omission or error that prevents E.
I	Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient.

[STAR] THE PROBLEM — annotated

A district checks an emergency-alert system. Let p be the proportion of alerts that arrive late. It tests $H_0 : p = 0.05$ versus $H_a : p > 0.05$, replacing the system if it rejects H_0 . Both plans use $n = 200$. The Safety office wants to avoid keeping a system with a 10% late rate. The Budget office wants to avoid replacing a system with a 5% late rate. The offices have not agreed how to compare these costs.

Dev: “Use A. Smaller α lowers both error risks.”

Lin: “Use B. Its higher power has no cost when safety matters.”

Test plans ($n = 200$)

Plan	α	Power at $p = 0.10$
A	0.010	0.748
B	0.050	0.877

FIRST TAKE — one sentence before you work the parts

Which plan would you try first? Name the office you prioritized and one competing cost.

Key: Not graded. Either plan needs a conditional defense; neither is cost-free.

[BOOK] ERROR PROBABILITIES TRADE OFF (keep on the page)

A Type I error rejects a true H_0 , and its probability is α . A Type II error fails to reject a false H_0 ; $\beta = 1 - \text{power}$. With sample size fixed, increasing α generally raises power and lowers β , but it also raises the false-positive risk. Increasing sample size can raise power while keeping the chosen α fixed. Choose a plan by naming the real consequence of each error.

DOK 1 (a) Identify what the table’s power at $p = 0.10$ means in this setting.

Key: Power is the probability that the test rejects H_0 and replaces the system when its true late-alert rate is 10%.

DOK 2 (b) Calculate the Type II error probability β for each plan. State each plan’s Type I error probability.

Key: A: $\beta = 1 - 0.748 = 0.252$ and Type I risk $\alpha = 0.010$. B: $\beta = 1 - 0.877 = 0.123$ and Type I risk $\alpha = 0.050$.

DOK 3 (c) In 4–5 sentences, recommend a plan for one office using both error risks. Explain the consequence of each error and what Dev and Lin each overlook. State why the other office might choose differently.

Key: For Safety, choose B: its chance of keeping a system with a 10% late rate is 0.123 versus 0.252 for A. But B has a 0.050 chance of unnecessary replacement when the rate is 5%, compared with 0.010 for A. Dev overlooks that A’s lower Type I risk comes with a higher Type II risk. Lin overlooks B’s higher replacement risk. Budget could reasonably choose A because it prioritizes avoiding unnecessary replacement; neither office’s priority makes one plan universally best.